

Sequences

MATH 146 – Calculus II

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Sequences

A **sequence** is an infinite list of numbers. More formally, if $\mathbb{N} = \{1, 2, 3, 4, \dots\}$ denotes the set of counting (or “natural”) numbers, then a sequence is a function $a : \mathbb{N} \rightarrow \mathbb{R}$. The n^{th} term of a sequence is usually denoted by a_n . (Not $a(n)$.)

Example: Sequences

There are lots of different notations that can be used to denote a sequence:

- $\left\{ \frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots \right\}$

- $\{7n + 1\}_{n=1}^{\infty}$

- $a_n = \frac{1}{2^n}$

- $a_n = \frac{(-1)^n}{n}$

- $a_0 = 1$ and $a_n = \frac{1}{n}a_{n-1}$

Sequences and Functions

There is *often* a strong relationship between sequences a_n and function $f(x)$ on the domain $[1, \infty)$.

Example: Consider the function $f(x) = \frac{1}{x}$.

Note: This isn't always the case. Consider $a_n = \frac{(-1)^n}{n}$.

Convergent and Divergent Sequences

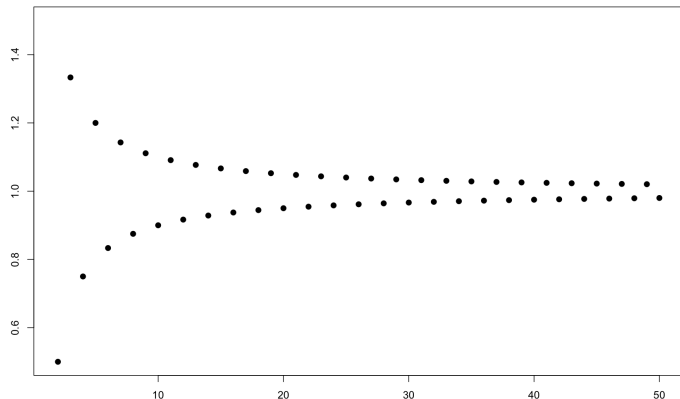
Just like we can talk about $\lim_{x \rightarrow \infty} f(x)$, we can also talk about $\lim_{n \rightarrow \infty} a_n$.

Convergent and Divergent Sequences

If the terms of $\{a_n\}$ approach a number L as n approaches ∞ , we say $\lim_{n \rightarrow \infty} a_n = L$ exists, or “ n **converges** to L ”. Otherwise, a_n **diverges**.

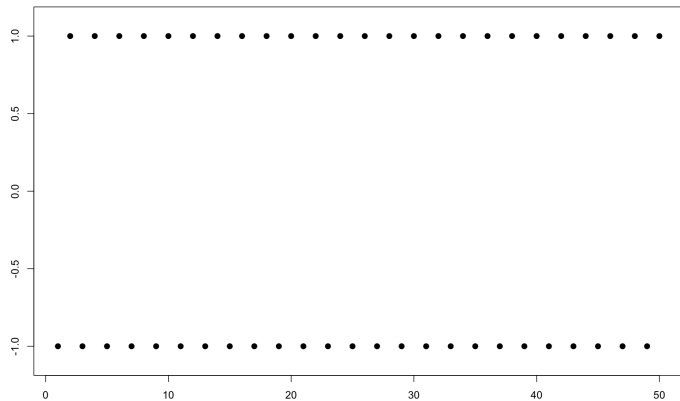
Example: Convergent and Divergent Sequences

Example: Is $a_n = 1 - \frac{(-1)^n}{n}$ convergent?



Example: Convergent and Divergent Sequences

Example: Is $a_n = (-1)^n$ convergent?



Example: Convergent and Divergent Sequences

Example: Is $a_n = 2^n$ convergent?

Limit Theorems

Most of the limit theorems we discussed carry over to sequences as well.

Theorem

Suppose that $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$ and that c is a constant.

Then:

- $\lim_{n \rightarrow \infty} (a_n + b_n) = A + B.$
- $\lim_{n \rightarrow \infty} (a_n b_n) = AB.$
- $\lim_{n \rightarrow \infty} (c \cdot a_n) = cA.$
- $\lim_{n \rightarrow \infty} \left(\frac{a_n}{b_n} \right) = \frac{A}{B},$ if $B \neq 0.$

Limit Theorems

Theorem

If $f(n) = a_n$ and $\lim_{x \rightarrow \infty} f(x) = L$, then $\lim_{n \rightarrow \infty} a_n = L$.

Example: Find $\lim_{n \rightarrow \infty} \frac{14n^4}{3n^4 - 7n^2 + 2}$.

Example: Find $\lim_{n \rightarrow \infty} \left(\frac{5+n}{n} \right)^n$.

Squeeze Theorems

Squeeze Theorem

If $a_n \leq c_n \leq b_n$ and $\lim_{n \rightarrow \infty} a_n = L$ and $\lim_{n \rightarrow \infty} b_n = L$, then $\lim_{n \rightarrow \infty} c_n = L$.

Example: Find $\lim_{n \rightarrow \infty} \frac{(-1)^n}{n}$.

Boundedness and Monotonicity

Boundedness

A sequence a_n is **bounded** if there is a constant M such that for *every* n , $|a_n| \leq M$.

Monotonicity

A sequence a_n is **monotonic** if $a_n \leq a_{n+1}$ for all n or $a_n \geq a_{n+1}$ for all n .

Theorem

Any bounded and monotonic sequence is convergent.

Limits of Recursive Sequences

Example: What, if anything, does the sequence given recursively by $a_0 = 5$ and $a_{n+1} = \frac{1}{2}a_n + 2$ converge to?